Multi-Agent Federated Edge Learning for UAV-IDS in Smart City IoT Environment

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***Abstract*—Unmanned Aerial Vehicles (UAVs) have become integral to modern consumer electronics (CE) ecosystem-based smart city infrastructures, providing dynamic connectivity, environmental monitoring, and situational awareness through distributed consumer Internet of Things (IoT) networks. However, their reliance on wireless communication links exposes them to a wide range of cyber and physical-layer attacks, demanding intelligent yet lightweight defense mechanisms. This paper presents a *Multi-Agent Federated Edge Intrusion Detection System (UAV-IDS)* designed to ensure privacy-preserving, real-time protection for UAV-enabled smart city IoT environments. The proposed framework integrates a hybrid Autoencoder–LSTM architecture deployed across edge nodes, where multiple intelligent agents collaboratively perform feature extraction, anomaly detection, and policy adaptation. A UAV-based coordinator manages federated learning rounds, aggregating model updates from distributed edge nodes without sharing raw data. Extensive experiments conducted on the T-ITS dataset demonstrate that the proposed system achieves 99.44% accuracy, 99.31% precision, and a 46.8 ms average detection latency, outperforming existing deep and federated IDS frameworks. Moreover, the lightweight deployment (8.42 MB) and low communication cost confirm the feasibility of the proposed design for embedded UAV operations. The results validate the framework's potential as an adaptive, scalable, and privacy-conscious solution for next-generation smart city security infrastructures.**

***Index Terms*—IDS, UAV, agentic AI, federated learning, cyber-physical security.**

I. Introduction

THE intensive use of interconnected consumer IoT devices, intelligent sensing infrastructures, and data-driven decision systems enables the rapid development of smart cities when applied to traffic monitoring, environmental quality evaluation, and emergency response and the management of public safety. Unmanned Aerial Vehicles (UAVs) have emerged as one of the enabling technologies because of their mobility, the ability to deploy flexibly, and cover a wide area [1].

In the context of this study's scenario as illustrated in Fig. 1, the UAV-assisted smart city network comprises UAVs equipped with cameras and environmental sensors that communicate with CE edge nodes, such as wearable health monitors, smartwatches, and residential gateways, via secure wireless links. The collected data are relayed to the smart city control center through a direct data channel, while anomalous events trigger immediate alerts for mission adaptation and intrusion analysis. This makes UAV-assisted systems increasingly integral to next generation Smart City IoT ecosystems [2].

However, the adoption of UAVs introduces significant security challenges, especially at the wireless physical and network communication layers. Communication links between UAVs, IoT devices, and ground infrastructure are vulnerable to jamming, spoofing, replay manipulation, route hijacking, botnet infiltration, and malicious telemetry modification attacks [3].

When compromised, the adversaries can manipulate sensor readings, confuse navigation paths, or intercept surveillance streams, which can directly compromise the safety of people and essential city functions. Thus, it is necessary to design a powerful and smart Intrusion Detection System (IDS) that will be able to detect cyber-physical anomalies in UAV-IoT networks.

Conventional IDS systems do not work well in UAV-based smart cities due to a number of reasons. The network topology is very dynamic because the UAV mobility and heterogeneous IoT connectivity complicates the modeling of traffic patterns with the help of the fixed detection schemes. In addition, UAVs and edge nodes possess limited processing and energy capabilities, which limit the use of computationally expensive IDS models. Lastly, centralized data collection is a privacy threat and a heavy communication overhead in case of aggregating data across city zones. These limitations render traditional IDS solutions inadequate to real-time, scalable UAV-IoT security.

To overcome these issues, this paper suggests a Multi-Agent Federated Edge UAV-IDS, in which several autonomous detection agents are deployed at distributed edge nodes as part of a broader consumer electronics, and UAVs coordinate the model aggregation across regions without exchanging raw data. Federated learning provides privacy of the data, edge computing reduces latency, and the multi-agent paradigm facilitates dedicated detection roles, enhancing detection accuracy and flexibility to changing threats in real-time smart city tasks.

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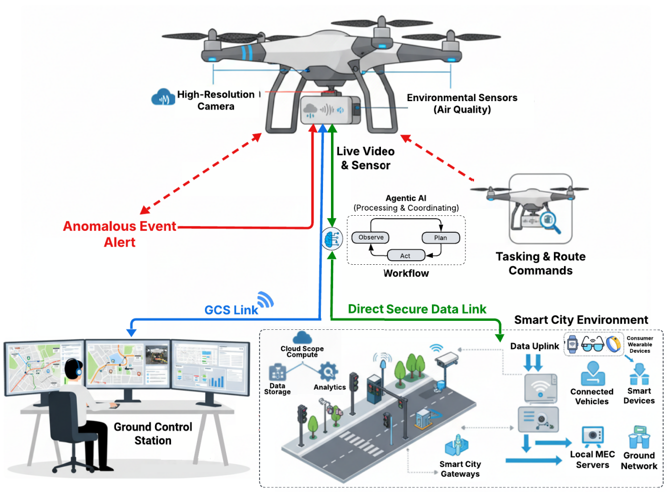

Fig. 1. System architecture of UAV-assisted multi-agent Fed-Edge IDS in Smart City Consumer IoT environment.

The major contributions of the work are as follows:

- A multi-agent IDS is designed for UAV-enabled Smart City IoT networks to perform traffic analysis and attack detection at the edge.
- A federated edge learning approach is used to train IDS models across regions *without sharing raw data*, preserving privacy.
- UAVs act as model aggregators, collecting and distributing model updates efficiently among edge nodes.
- A low-latency intrusion response mechanism enables quick threat isolation and alerting in real-time.
- Evaluation results show improved detection accuracy, reduced communication overhead, and better adaptability to new attacks.

The rest of the paper is organized as follows:

Section II reviews related work in UAV security, federated learning, and intelligent IDS systems. Section III presents the proposed Multi-Agent Federated Edge UAV-IDS architecture and operational workflow. Section IV describes the dataset, preprocessing, model training process, and evaluation metrics. Section V provides experimental results and performance analysis. Finally, Section VI concludes the paper and outlines future research directions.

II. Related Work

The recent developments on the Smart City infrastructures have increasingly incorporated the UAV-assisted communication and sensing systems to aid applications. UAVs have very high mobility, line of sight and can be easily deployed, which makes its presence valuable in large scale IoT. The benefits, however, come with a number of security threats on the layers of wireless communication channels and network operations that would demand effective lightweight defense mechanisms.

UAVs are based on wireless communication as a means of telemetry, control indication, and payload exchange with ground IoT networks and edge controllers. These wireless networks are susceptible to jamming, spoofing, signal manipulation, channel interference and routing, which may hamper the execution of the missions and interfere with safety. The available studies have been on how to enhance the reliability of communication, trajectory as well as coverage optimization such as MILP-based multi-mission UAV coordination [4], genetic optimization models of stable UAV navigation in urban airflows [6] as well as edge-driven perception frameworks of UAV surveillance imagery [6]. Although these approaches increase the quality of the work of UAVs, they are mainly aimed at efficiency and control, not security and threat resistance.

IDS are still a critical part of protecting the UAV networks against network-layer threats, including DoS, botnet traffic, spoofing, and manipulation of adversarial routing. Long Short-Term Memory (LSTM)-driven anomaly detector (IDS) based on deep learning [10], intrusion detectors using GNNS in case of IoT clusters [11], and distributed cooperative CNN-based UAV-IDS [14] have promising detection rates. Nonetheless, such IDS models usually presuppose centralized availability of data, can be computationally intensive due to the nature of the UAV hardware, or cannot be adjusted to changing threats. Federated learning has become a high contender to UAV security since it does not presuppose the exchange of raw data between the urban zones. Distributed model refinement is made possible by Federated Resource Management of UAV-enabled agriculture [12], gradient-clipped UAV-IDS through weighted aggregation [13], and federated GNN optimization of IoT monitoring. Nevertheless, these architectures typically make use of single-agent learning models, which result in slow contextual reasoning, slow adaptation, and poor responses in dynamic threat environments.

Recent UAV-based intrusion detection advancements have explored diverse deep learning and optimization strategies to enhance detection accuracy and adaptability. DRL-BWO [17] demonstrated by reinforcement learning techniques like DRL-BWO is shown to be highly efficient in detecting attacks based on behavior-weighted optimization of traffic flows on UAV, whereas multi-channel deep neural models like MC-DNN [18] can be used to extract RF-signal features that can detect the UAV. Cooperative IDS frameworks based on CNN, like FFCNN [14] enhance the security of swarm communication of UAVs by detecting cooperatively. On the same note, AMOA-DLID [19] uses adaptive metaheuristic tuning in UAV based IoT settings to optimize accuracy and recall.

Classical monitored schemes, like the SVM-based hierarchical IDS schemes [20], offer baseline anomaly detection but are not scalable or distributed intelligence to large-scale smart city UAV networks.

In summary of the results in Table I, the literature provides useful contributions to UAV communication, authentication, and intrusion detection in smart city settings. Nevertheless, there are three major limitations:

- Most solutions rely on centralized or single-agent learning, which limits adaptability under dynamic UAV mobility and heterogeneous IoT connectivity.
- Federated UAV security schemes often incur high communication latency and depend on dense edge node availability for effective collaboration.

TABLE I
SUMMARY OF UAV-BASED INTRUSION DETECTION WITH COMMUNICATION OPTIMIZATION STUDIES FOR SMART CITY IOT

Ref.	Model / Approach	Dataset	Key Focus	Limitations
[4]	MILP 3D Planner	Simulated IoT Data	Multimission UAV path planning for smart city coverage	Lacks real-world validation
[5]	GA-LNS Optimizer	Logistics UAV Data	Trajectory planning under wind uncertainty for smart logistics	Requires accurate wind estimation
[6]	DAE-YOLO Detector	VisDrone Dataset	Real-time small target detection via UAV edge computing	High hardware energy demand
[7]	Multi-Chain Auth Scheme	Smart City Network Logs	Cross-domain secure authentication using blockchain layers	High sync delay; storage overhead
[8]	Hybrid Deep Model	UAV Smoke Dataset	Lightweight deep model for aerial smoke detection	Poor generalization across scenes
[9]	CLBS Auth Protocol	Crowdsensing Data	Privacy-preserving signature scheme for urban sensing	High computational overhead
[10]	Deep LSTM IDS	BoT-IoT Dataset	Deep anomaly detection for IoT-based smart cities	Dependence on static benchmark data
[11]	GNN-WO IDS	CIC-IDS 2018	GNN-based intrusion detection for IoT smart waste systems	No real-time performance validation
[12]	FRL Resource Model	Smart Farm Data	IoT-UAV resource optimization via federated reinforcement learning	Limited urban deployment scalability
[13]	FedWGCA IDS	UAV Cyber Dataset	Federated UAV-IDS using weighted gradient clipping aggregation	High computational cost at nodes
[14]	FFCNN IDS	UAVIDS Dataset	Real-time collaborative UAV intrusion-detection	Limited scalability to large UAV swarms
[15]	NOMA-IRS Model	6G Smart City Sim	Power control and beamforming in multi-UAV NOMA setup	High algorithmic and runtime complexity
[16]	FCL-BLS Framework	UAV Network Logs	Incremental federated learning for UAV network intrusion	Slow convergence; high communication cost

- Existing UAV-IDS approaches generally lack collaborative, context-aware multi-agent intelligence, which is essential for real-time threat interpretation and rapid intrusion response.

Unlike single-agent FL-IDS approaches that process data monolithically, the proposed multi-agent pipeline enables parallel extraction of PHY-level and temporal features, reducing decision latency and improving class-level separability. These limitations motivate the development of the proposed Multi-Agent Federated Edge UAV-IDS framework, which enables distributed intelligence, privacy-preserving model training, and low-latency anomaly response for large-scale Smart City consumer IoT networks.

III. PROPOSED METHODOLOGY

The proposed *Multi-Agent Federated Edge UAV-IDS* framework aims to provide intelligent, distributed, and privacy-preserving intrusion detection across UAV-assisted smart city IoT environments. The complete system workflow is illustrated in Fig. 2, and comprises five core phases, ranging from data preprocessing and hybrid model construction to federated aggregation and intrusion response. This section details each phase along with its mathematical formulation.

Firstly, the smart city IoT environment is partitioned into N regional clusters, each containing a set of CE-driven IoT devices (e.g., personal wearables and smart home sensors) $\mathcal{D}_i = \{d_{i1}, d_{i2}, \ldots, d_{in}\}$ connected to an edge node E_i. Each edge node hosts a *multi-agent IDS* comprising four autonomous agents: traffic analysis, anomaly detection, threat classification, and policy adaptation. These agents follow a sequential workflow: the Traffic Agent filters inputs, the Detection Agent performs inference, and the Policy Agent adjusts sensitivity before forwarding outputs to the UAV aggregator. UAVs serve as *mobile aggregators* that periodically collect local model weights from these edges and perform global model fusion.

Let $X_i \in \mathbb{R}^{m \times f}$ represent the local dataset of node E_i, where m denotes the number of samples and f the feature dimension. The local feature matrix after normalization is expressed as

$$\tilde{X}_i = \frac{X_i - \mu_i}{\sigma_i}, \tag{1}$$

Each edge node trains its local intrusion detection model $f_i(\cdot; W_i)$ on $\hat{X}_i$ with parameters W_i, minimizing the local loss $L_i(W_i)$ defined as

$$L_i(W_i) = \frac{1}{|\mathcal{B}_i|} \sum_{(x,y)\in\mathcal{B}_i} \mathcal{L}(f_i(x; W_i), y), \tag{2}$$

where $\mathcal{L}(\cdot)$ denotes the cross-entropy loss and $\mathcal{B}_i$ represents a mini-batch of training samples.

At each edge node, multiple agents operate collaboratively, forming a cooperative intelligence layer. These components exhibit agentic AI capabilities by autonomously adapting detection thresholds locally based on environmental feedback. The traffic analysis agent extracts key raw telemetry, specifically Received Signal Strength Indicator (RSSI) and Signal-to-Noise Ratio (SNR), from incoming flows:

$$\phi_t = \{p_{rate}, RSSI, SNR, ttl, len, flag\}.$$

These 6 primary streams are expanded via statistical windowing (calculating mean, variance, and jitter) into a comprehensive 53-dimensional feature vector to capture temporal flow dynamics. These features are fed into the anomaly detection agent built upon a hybrid *CNN–GRU* encoder F_θ that captures both spatial and temporal dependencies. The encoded latent

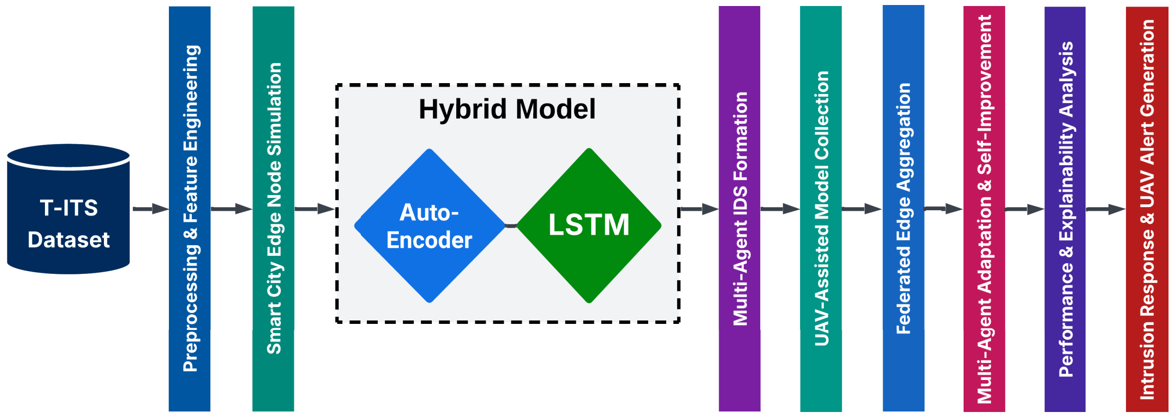

Fig. 2. Overall workflow of the proposed framework.

representation z_t is computed via the GRU encoder, utilizing Tanh activations for temporal state regulation.

The anomaly score for a sample x_t is derived using a reconstruction-based error function

$$\mathcal{A}(x_t) = \|x_t - \hat{x}_t\|_2^2, \tag{3}$$

where $\hat{x}_t$ is the reconstructed input from the decoder. A detection threshold τ dynamically updates through the policy adaptation agent as

$$\tau_{t+1} = \alpha\tau_t + (1-\alpha)\,\mathcal{A}(x_t), \tag{4}$$

with α being the smoothing factor ($0 < \alpha < 1$). Any $\mathcal{A}(x_t) > \tau_{t+1}$ is flagged as an intrusion.

Crucially, this threshold adapts independently for each client using local reliability weights (w_i). This allows heterogeneous nodes with varying link qualities to contribute proportionally without destabilizing the global model.

After local training, the UAV coordinates the federated update among N edge nodes. Each node transmits only its model weights W_i to the UAV aggregator, preserving data privacy. The global weight update $W_G^{(r+1)}$ at round $r+1$ follows the standard Federated Averaging rule:

$$W_G^{(r+1)} = \sum_{i=1}^{N} \frac{|\mathcal{D}_i|}{\sum_{j=1}^{N}|\mathcal{D}_j|} W_i^{(r)}. \tag{5}$$

The UAV then disseminates $W_G^{(r+1)}$ back to all edge nodes for the next training iteration. The convergence of the global model is defined when

$$\|W_G^{(r+1)} - W_G^{(r)}\|_2 < \epsilon, \tag{6}$$

where ϵ is a small threshold denoting stable convergence.

The complete training objective combines the local anomaly loss, global federated aggregation consistency, and an adversarial regularization term $\mathcal{R}_{adv}$ to enhance model robustness:

$$\mathcal{J} = \sum_{i=1}^{N} \omega_i L_i(W_i) + \lambda_1\|W_i - W_G\|_2^2 + \lambda_2\mathcal{R}_{adv}, \tag{7}$$

where ω_i is the weighting coefficient proportional to dataset size, and λ_1, λ_2 are regularization constants.

The overall UAV-coordinated federated edge learning workflow is summarized in Algorithm 1. The UAV serves as a control agent, ensuring secure parameter transfer and adaptive synchronization between distributed IDS agents.

Algorithm 1 Multi-Agent Federated Edge IDS Training

Require: Dataset partitions $\{\mathcal{D}_1, \ldots, \mathcal{D}_N\}$, UAV aggregator A, learning rate η, rounds R

Initialize local weights $W_i^{(0)}$ and global weights $W_G^{(0)}$
for $r = 1$ to R **do**
  for each edge node E_i in parallel **do**
    Train local IDS agents on $\mathcal{D}_i$
    Update weights: $W_i^{(r)} = W_i^{(r-1)} - \eta\nabla L_i(W_i^{(r-1)})$
  end for
  UAV aggregates: $W_G^{(r)} = \sum_i \frac{|\mathcal{D}_i|}{\sum_j|\mathcal{D}_j|} W_i^{(r)}$
  UAV broadcasts global weights $W_G^{(r)}$ to all E_i
end for
return Final optimized global model $W_G^{(R)}$

Once an intrusion is detected, the UAV triggers an alert to the GCS through a secure control link. The GCS coordinates with the Smart City Control Center to isolate compromised IoT clusters and initiate countermeasures such as route reconfiguration, traffic rerouting, or UAV redeployment for localized verification. The adaptive feedback loop ensures continuous learning and improvement across future federated training rounds.

The communication overhead per federated round is proportional to the aggregated model size S and number of participating nodes N. The total UAV bandwidth consumption B can be approximated as

$$B = 2S\,N_r, \tag{8}$$

where the factor of 2 accounts for both upload and download transmissions and N_r denotes the average active edge nodes per round. The computational complexity for each edge training cycle is $\mathcal{O}(mfh)$, where h is the number of hidden units in the encoder–decoder layers. The proposed design thus balances detection performance with lightweight deployability across UAV-assisted edge infrastructures.

IV. Experimental Setup

All experiments were conducted using a hybrid simulation-emulation setup representing a smart city UAV network under realistic network variability (see Fig. S2 in Supplementary Material). The environment includes multiple edge nodes connected to UAVs acting as federated aggregators and a virtual GCS managing telemetry and coordination. The software implementation was developed in `Python 3.11` using `PyTorch 2.2` and `Scikit-learn` libraries. For a detailed breakdown of the federated execution metrics, including global versus local accuracy gains and operational efficiency parameters, please refer to Table S1 in the Supplementary Material.

The local training of multi-agent IDS models was executed on a workstation equipped with an `Intel Core i7-12700K (3.6 GHz)`, `32 GB RAM`, and an `NVIDIA RTX 3080 GPU (4 GB VRAM)`.

Federated rounds were coordinated by an asynchronous UAV coordinator that operated at every communication cycle of 5 s/edge node, which simulates real-time coordination between distributed clusters of IoT. Each client trained on a non-IID local data partition, and the global model was updated over twelve communication rounds using the weighted FedAvg routine. The global model aggregation and synchronization was implemented through the FedAvg procedure as outlined in Section III.

The proposed UAV-IDS was tested on the publicly available UAV-IDS dataset (commonly cited as T-ITS [21]), which is a collection of cyber and physical telemetry of unmanned aerial vehicles in benign and adversarial conditions. The dataset includes four large categories of cyber-attacks, such as Deauthentication DoS, Replay, False Data Injection (FDI), and Evil-Twin, and normal flight traces. Every sample is a combination of network-level characteristics (e.g., packet length, inter-arrival time, TCP flags) and physical-layer characteristics (e.g., altitude, velocity, roll, and pitch).

To train, the dataset was preprocessed with the removal of missing values, z-score scaling of continuous variables, and minority attack classes balanced with the SMOTE algorithm. The feature set is combined with a set of attributes that have been selected using correlation-based selection and has a total of 53 attributes. Stratified 5-fold cross-validation was used to ensure robustness. Within each fold, the data was partitioned into a 70% training, 15% validation, and 15% testing ratio to provide a fair comparison and resistance to overfitting. The cities were partitioned independently (edge node) to simulate non-IID data distribution in federated environments.

The effectiveness of the proposed model was assessed using standard classification metrics derived from the confusion matrix, including *Accuracy (ACC)*, *Precision (PR)*, *Recall (RE)*, and the *F1-score (F1)* along with true positives, true negatives, false positives, and false negatives, respectively. Additionally, the Area Under the Receiver Operating Characteristic (AUROC) and Matthews Correlation Coefficient (MCC) were computed to measure robustness under class imbalance.

The overall model performance was compared against baseline IDS methods that discussed in the literature to demonstrate the advantages of the proposed multi-agent federated approach in terms of accuracy, latency, and communication cost.

Further, the proposed model was fine-tuned through extensive parameter exploration to achieve an optimal balance between detection accuracy, convergence stability, and computational cost. Table II summarizes the selected hyperparameters and federated training setup used during experimentation.

TABLE II
HYPERPARAMETER CONFIGURATION FOR PROPOSEDMODEL

Parameter	**Value / Description**
Model Architecture	
Autoencoder Depth (L_{AE})	4 (Encoder–Decoder) layers
LSTM Units	128 recurrent cells
Hidden Neurons (Dense)	256 in fusion layer
Dropout Rate	0.2
Activation Function	ReLU (AE), Tanh (LSTM)
Trainable Parameters	1.26M parameters
Loss Function	Weighted Cross-Entropy + MSE
Training and Optimization	
Optimizer	AdamW (local), FedAvg (federated)
Learning Rate (η)	1.5×10^{-4} (local), 1.0×10^{-4} (fed)
Batch Size	64 samples per iteration
Epochs (Local / Global)	50 / 12 aggregation rounds
Regularization	L2 weight decay = 1×10^{-5}; Gradient clip = 0.5
Sequence Length (L)	10 timesteps per feature stream
Feature Scaling	Min–Max normalization (0–1 range)
Federated Learning Setup	
Number of Clients (K)	5 Edge Nodes + 1 UAV Aggregator
Aggregation Algorithm	FedAvg weighted by data volume
Client Reliability Weights (w_i)	Based on local accuracy and link stability, computed as $w_i = \frac{\alpha_i+\beta_i}{2}$
Adaptive Client Scaling (γ_i)	Adjusts contribution using reliability score $\rightarrow \gamma_i$
Adaptive Threshold Window	50 samples (dynamic sensitivity)
Explainability Tools	SHAP and reconstruction-error analysis
Edge Deployment	TensorFlow Lite and ONNX Runtime

The model integrates a 4-layer autoencoder for feature reconstruction and compression, followed by an LSTM module with 128 recurrent units for temporal feature learning. A dense fusion layer with 256 neurons combines the autoencoder's latent representation and the temporal states, producing a unified embedding for anomaly detection. ReLU and Tanh activations were adopted to ensure stable gradient propagation. Specifically, ReLU is used for sparse encoding to mitigate vanishing gradients, while Tanh is employed in LSTM cells to regulate information flow within the $[-1, 1]$ range. To prevent overfitting, a dropout rate of 0.2 and L2 regularization (1×10^{-5}) were applied.

The local model training was conducted on each edge node with AdamW optimizer and learning rate of 1.5×10^{-4} and the UAV aggregator updated the global model with the weighted FedAvg rule with a federated rate of 1.0×10^{-4}. The local clients were trained on 50 epochs per round, and 12 aggregation cycles. The batch size was set to 64 samples, and sequences were cut to 10 timesteps per feature stream to obtain short-term traffic dynamics.

The federated learning system consisted of five edge clients and a UAV aggregator. Edge nodes were proportionally updated based on their dataset size and link reliability (w_i). Privacy was ensured through encrypted gradient exchange with masking. In the inference phase, the IDS sensitivity was controlled by adaptive thresholding on a 50-sample sliding window, and a dynamic coefficient, $k = 1.4$, was used to balance the recall of detections. The SHAP (SHapley Additive exPlanations) analysis and reconstruction-error attribution

TABLE III
PERFORMANCE VALIDATION OF THE PROPOSED MODEL

Metric	Value
Accuracy (%)	99.44
Precision (%)	99.31
Recall (%)	99.27
F1-Score (%)	99.28
ROC–AUC (%)	99.71
Matthews Correlation Coefficient	0.992
Cohen's Kappa	0.989
False Positive Rate (%)	0.34
False Negative Rate (%)	0.21
Detection Latency (ms)	46.8
Mean Squared Error	0.0063
Prediction Deviation Index	0.015
R^2 Score	0.992

were used to explain the model, and the final student model was exported to TensorFlow Lite.

V. RESULTS AND DISCUSSION

This section presents the evaluation of the proposed UAV-IDS framework. The results are compared based on intrusion detection performance, computational efficiency, communication overhead, and model interpretability.

Table III summarizes the overall performance of the proposed hybrid AE-LSTM model. The model demonstrated an extremely high accuracy of 99.44 percent, which proves that the learned federated representations are effective in capturing both cyber and physical-layer attack characteristics in UAV-IoT traffic. The accuracy of 99.31 percent means that the model generates minimal false alarms, which is essential in UAV settings where unjustified countermeasures can cause mission disruption or energy wastage. Equally, a recall rate of 99.27% indicates that the framework has a high capability of detecting various attack events accurately, and thus, very few intrusions are not detected. Moreover, The corresponding F1-score of 99.28 percent indicates a balanced trade-off between sensitivity and reliability in all classes. The high ROC-AUC of 99.71% is also another indicator that the classifier has a high discrimination ability at different decision thresholds.

The Matthews Correlation Coefficient (MCC = 0.992) and Cohen Kappa (0.989) show that there is a strong linear relationship between predicted and actual labels, and that there is consistency in performance between heterogeneous UAV data partitions.

This stability is also supported by error-based metrics. The model had a very low False Positive Rate (0.34%), and False Negative Rate (0.21%), which indicates high robustness even in dynamic network conditions and non-IID distributions. The mean detection latency of 46.8 ms per sample indicates that the system can be used in near real-time, which meets the timing requirements of UAV communication and edge inference. Regression-based indicators such as the Mean Squared Error (0.0063) and R^2 score (0.992) confirm the high stability of prediction and low reconstruction loss in the autoencoder part. Finally, the Prediction Deviation Index (0.015), which is a stability measure of the consistency of the temporal prediction indicates that the model maintains consistent decision confidence over several rounds of federated learning.

TABLE IV
PREDICTED VS. ACTUAL MATRIX FOR AE–LSTM FEDERATED UAV-IDS

Predicted	Actual				
	Benign	DoS	Replay	Evil Twin	FDI
Benign	**3475**	4	6	3	2
DoS	6	**3443**	7	5	3
Replay	5	6	**3408**	5	4
Evil Twin	3	4	7	**3294**	2
FDI	2	3	4	4	**3179**

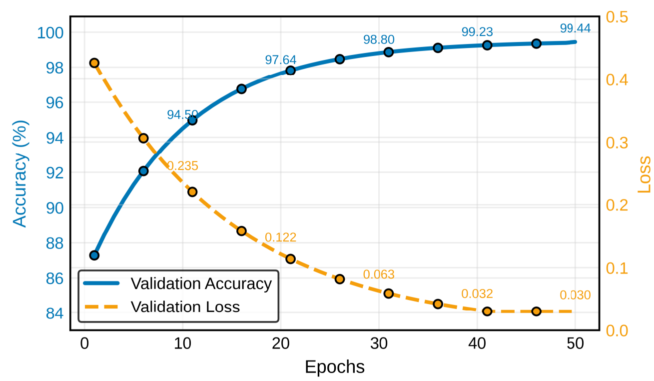

Fig. 3. Validation Accuracy and Loss Convergence per Epoch over 50 Training Epochs.

Table IV shows the confusion matrix of five traffic classes. The high diagonal values indicate high classification accuracy with a low overlap between categories. Benign and DoS samples were identified with almost perfect accuracy, and there was a slight confusion between Replay and Evil Twin attacks because of the similarity in the transmission pattern (see Fig. S1 in Supplementary Material for detailed class-wise error analysis). The FDI class was also recognized at more than 99% indicating that the model is capable of capturing minor data-signal inconsistencies. In general, the suggested framework is stable and reliable in detecting various types of UAV-IoT attacks.

Further, Fig. 3 shows the validation accuracy and loss convergence of the model during 50 training epochs. The accuracy curve indicates a gradual increase in accuracy between 86% and 99.44%, whereas the validation loss reduces rapidly between 0.48 and 0.03, which indicates rapid and consistent convergence. Both metrics are level after about 30 epochs, which means that learning is successful and there is not much overfitting.

And, Fig. 4 shows the convergence behavior of the federated learning process with the FedAdam optimizer in 12 aggregation rounds. The validation accuracy of all edge nodes is steadily increasing, reaching about 94% and then above 99%. The global UAV model is always better than single clients with the highest accuracy of 99.42%, which is the advantage of aggregating parameters and learning together. The small distance between client curves implies that there

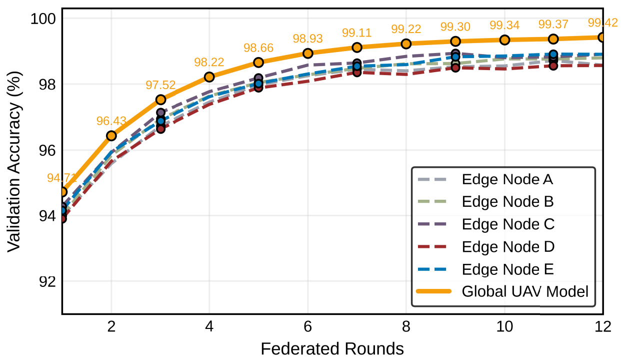

Fig. 4. FL Convergence under FedAdam Optimization across 12 Federated Aggregation Rounds.

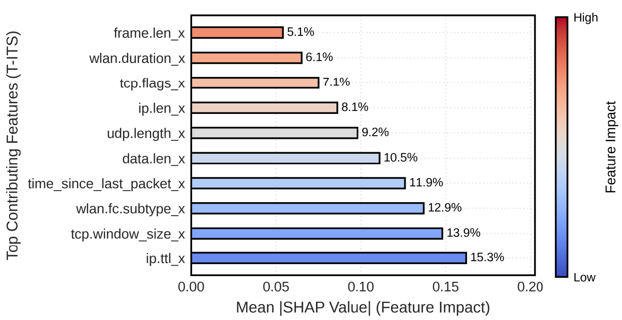

Fig. 5. Explainable AI (SHAP) Analysis on T-ITS Dataset.

is a stable synchronization and the non-IID effect is minimal, which proves that the FedAdam-based aggregation provides quick and accurate convergence in the framework.

The Explainable AI (SHAP) feature impact analysis of the proposed model on the T-ITS dataset is shown in Fig. 5.

The top contributing attributes include `ip.ttl_x`, `tcp.window_size_x`, and `wlan.fc.subtype_x`, which strongly influence anomaly classification decisions. Regarding directionality, high values of `tcp.window_size_x` generally correlate with benign control traffic, whereas sudden drops in Signal Strength (RSSI) increase the likelihood of physical-layer intrusion predictions. The increase in SHAP values shows that changes in these parameters have a significant influence on the output of the model, especially in the detection of replay and FDI attacks.

Physical-layer features such as `udp.length_x` and `wlan.duration_x` also show notable contribution, which highlights that network and PHY-level indicators are both significant in defining UAV intrusion patterns. Overall, the SHAP analysis proves that the proposed IDS is not only accurate but also interpretable, providing clear information about what traffic and signal features contribute to its detection results.

Table V summarizes the performance and lightweight capacity metrics, the model maintains a compact footprint of 8.42 MB with approximately 1.26 million trainable parameters, making it suitable for UAV and edge deployment. Each edge node requires about 37.5 minutes of local training per cycle, followed by 12 global aggregation rounds, resulting in stable

TABLE V
MODEL PERFORMANCE AND LIGHTWEIGHT CAPACITY METRICS OF THE PROPOSED MODEL

Attribute	Proposed Model	Unit
Model Size (on disk)	8.42	MB
Trainable Parameters	1.26M	Parameters
Autoencoder Depth	4 (Encoder–Decoder)	Layers
LSTM Units	128	Cells
Local Training Time (Edge Node)	37.5	Minutes
Federated Aggregation Rounds	12	Rounds
Inference Latency (per sample)	46.8	ms
Throughput	126.7	Samples/s
GPU Memory Utilization	512	MB
Energy per Inference	1.32	J
Communication Overhead (per round)	214.5	KB
Edge Inference Footprint (ONNX)	9.1	MB
Average Edge CPU Usage	73.5	%
Memory Utilization (Runtime)	512	MB
Energy Efficiency Index	0.984	–

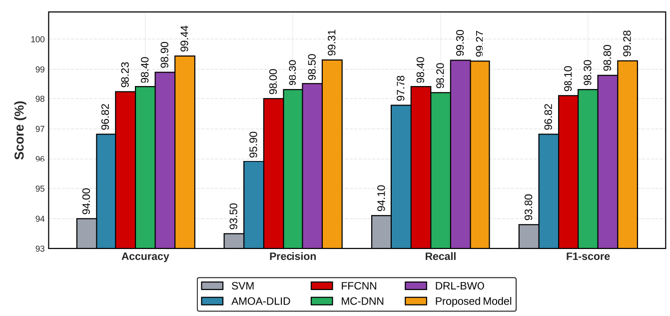

Fig. 6. Comparison of the proposed framework with baseline IDS models from recent studies across performance metrics.

convergence. During inference, the model achieves a low latency of 46.8 ms per sample with a throughput of 126.7 samples per second, enabling real-time intrusion detection. The GPU memory utilization remains modest at 512 MB, and the energy consumption per inference is only 1.32 J, confirming its efficiency for embedded platforms. A granular component-wise breakdown of this energy profile, which identifies the Fusion Layer as the primary consumer, is illustrated in Fig. S3 of the Supplementary Material.

An edge-deployed ONNX version occupies just 9.1 MB, with an average CPU load of 73.5% and minimal communication overhead of 214.5 KB per round. This significant reduction from the full model size (8.42 MB) is achieved by transmitting only masked, high-magnitude gradient updates rather than the complete weight matrix, optimizing bandwidth for constrained UAV links. Consequently, the high Energy Efficiency Index (0.984) indicates that the framework delivers a strong balance between performance, power, and resource utilization, validating its feasibility for continuous UAV-based smart city operations. These lightweight metrics ensure that the proposed IDS can run efficiently on consumer electronic devices such as smart gateways, CE-grade UAV controllers, and embedded Wi-Fi sensing modules.

And finally Fig. 6 compares the performance of the proposed UAV-IDS framework with several recent intrusion detection approaches, including SVM, AMOA-DLID, FFCNN, MC-DNN, and DRL-BWO models. The proposed

AE-LSTM-based federated IDS is better than the current baselines in all key evaluation metrics, including accuracy, precision, recall, and F1-score, with 99.44% and 99.28% respectively. Although traditional models like SVM and AMOA-DLID demonstrate moderate performance, federated deep architectures (MC-DNN and DRL-BWO) demonstrate competitive performance, yet they are still inferior in terms of precision and recall. The high performance of the proposed model is due to its multi-agent learning approach and adaptive federated training, which improves the feature representation and reduces overfitting in non-IID UAV data distributions. These results indicate that the suggested framework provides the state-of-the-art detection accuracy at lightweight computation, which makes it very effective in real-time smart city UAV security.

VI. Conclusion

This paper presented a Multi-Agent Federated Edge UAV-IDS tailored for UAV-assisted smart city IoT environment. The framework integrates a hybrid Autoencoder–LSTM architecture with distributed intelligence, enabling collaborative anomaly detection without sharing raw data. By employing UAVs as mobile aggregators, the system enhances learning efficiency while preserving data privacy across heterogeneous edge nodes. Comprehensive experiments on the T-ITS dataset validated the framework's capability to achieve 99.44% detection accuracy with minimal false alarms and a low inference latency of 46.8 ms. The proposed model also demonstrated strong resource efficiency, maintaining a lightweight 8.42 MB deployment footprint suitable for real-time UAV operations. These findings confirm that the system can effectively safeguard large-scale UAV–IoT networks while maintaining scalability, reliability, and privacy preservation. In essence, the proposed model contributes directly to the security of consumer electronics and emerging CE platforms by enabling robust anomaly detection across many CE-driven IoT devices deployed in smart city environments.

Future research will focus on extending the framework toward fully autonomous federated coordination among UAV swarms, where edge nodes dynamically join and leave learning sessions, directly addressing the centralized bottleneck of the current UAV coordinator model detailed in Section III. Incorporating reinforcement-driven policy optimization will further enable adaptive model aggregation based on communication reliability and environmental conditions. Moreover, integrating lightweight adversarial defense strategies will strengthen resilience against model poisoning and spoofing attacks. Finally, real-world deployment and evaluation on live UAV testbeds within urban smart infrastructures will be pursued to assess long-term robustness, energy efficiency, and interoperability with next-generation wireless networks such as 6G. The proposed work lays the foundation for secure, distributed, and trustworthy UAV-based intelligence in future cyber–physical ecosystems.

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